Lab 43 — Windows Security Configuration and BitLocker Encryption

> **Course:** CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)
> **Topic 07:** Security — Core 2, 28% of Core 2
> **Exam objective:** Configure Windows security settings including Defender, the firewall, UAC, account policy and BitLocker encryption (Core 2 objectives 2.5 and 2.6).

Goal

You configure and verify every Windows security setting the Core 2 objectives name, on a real machine, recording the command that checks each one — because in support work you need to verify state quickly, not click through settings pages.

What you'll produce

A verified Windows security configuration record with a check command for each setting.

Tools and equipment

Windows PC, Windows Security, Defender Firewall, gpedit, BitLocker, PowerShell as administrator

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Open Windows Security and record the status of every protection area: virus and threat protection, firewall, app and browser control, device security and account protection.

```
start windowsdefender:
```

2. Verify Defender Antivirus is enabled with current definitions, and record the definition version and date.

```
powershell -Command "Get-MpComputerStatus | Select-Object
AntivirusEnabled,RealTimeProtectionEnabled,AntivirusSignatureLastUpdated"
```

3. Record why definition currency matters: signature-based detection cannot identify a threat whose signature it does not yet hold.
4. Check the firewall state for all three profiles — domain, private and public — and record which are enabled.

```
netsh advfirewall show allprofiles state
```

5. Examine the inbound rules and record how a rule is scoped by program, port, protocol and profile.

```
netsh advfirewall firewall show rule name=all dir=in | findstr /C:"Rule Name" | head
-10
```

6. Record the default firewall posture: block inbound unless explicitly allowed, permit outbound, and add exceptions only for a stated business need.
7. Check User Account Control's configured level and record what each level prompts for.

```
reg query HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System /v
ConsentPromptBehaviorAdmin
```

8. Record the difference between running as a standard user and as an administrator, and why standard-user daily operation limits malware impact.
9. Examine the local account policy and record the password length, complexity, history and lockout threshold currently enforced.

```
net accounts
```

10. Record the local users and their group memberships, and identify every account holding administrative rights.

```
net user && net localgroup Administrators
```

11. Check BitLocker status on the system volume and record the protection state, the encryption method and the key protectors in use.

```
manage-bde -status C:
```

12. Record what BitLocker protects against — theft of the physical drive — and note that it requires a TPM, that the recovery key must be stored somewhere other than the encrypted machine, and that BitLocker To Go covers removable drives.

Test it — verification

Every check command returns output you have recorded; Defender is confirmed enabled with a definition date; all three firewall profiles have a recorded state; net accounts output is captured; and the

BitLocker record states where the recovery key is stored.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the `apt-get install` step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against `labs/tools.md`. All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete worksheet.md as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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